

Does Order Matter? Sequential Models for Spoken Swahili Word Recognition

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Abstract—Voice interfaces built on a handful of spoken words could serve users with limited literacy, but African languages such as Swahili have little labelled speech. We ask how effectively sequential models recognise 12 spoken Swahili words (the Zindi Swahili Audio Classification data, 4,200 clips), and what evidence supports their strengths and limitations. Five approaches differ in how they model temporal order: order-free MFCC statistics with logistic regression, a GMM-HMM, a BiLSTM with attention, a temporal convolutional network (TC-ResNet), and a fine-tuned multilingual self-supervised Transformer (XLS-R). All five share one frozen split and protocol, and the test split was used only after every configuration was frozen. The order-aware models reach 96–98% test accuracy, against 79.5% for order-free statistics, and a stress test that reverses or shuffles each word shows that they rely on order. The fine-tuned XLS-R is best (98.1% accuracy, log loss 0.093), and more accurate than every other model under Holm-corrected McNemar tests. With only 24 clips per word, it is about as accurate as the models trained from scratch on 245. The HMM, BiLSTM and TC-ResNet tie on accuracy, and the convolutional model has the lowest log loss of the three, at 150K parameters. Ten test clips defeat every order-aware model. Confident learning and an external recogniser trace several of them to the data: a likely label error, words said in English, and crops that missed the word.

Index Terms—Swahili, keyword spotting, sequence models, HMM, LSTM, temporal convolution, wav2vec 2.0, low-resource languages

Code and results: <https://github.com/Hassan-Adelani-Luqman/nlp-sequential-modeling-group21>

Demo video: <https://youtu.be/Gv1f1HYyZ18>

Contribution tracker: <https://docs.google.com/spreadsheets/d/11V5C-49tpxNkssPY9IWK0iY321QZCv3a-YPMJONuf9E/edit?usp=sharing>

I. INTRODUCTION

Spoken digits and a spoken “yes” or “no” are enough to run a mobile-money transfer, a health hotline or a phone menu. For users with limited literacy, voice is often the only practical interface [1]. Swahili is widely spoken across East Africa. Yet it remains a low-resource language for speech technology: it has only limited labelled data [2], and it was missing from the first releases of the largest open crowdsourced speech corpus [3]. Recognising a small Swahili vocabulary reliably, cheaply and with honest confidence is therefore a useful problem in its own right, as well as a test of how well modern sequence models transfer to an African language.

We study the Zindi Swahili Audio Classification challenge [4]: 4,200 crowdsourced clips of 12 words, the digits *moja* to *kumi* (one to ten), *ndio* (yes) and *hapana* (no), scored by log loss. A spoken word is a *sequence* of acoustic frames, and its identity depends on the order of its sounds.

The vocabulary makes this concrete: *sita* (six) and *tisa* (nine) contain the same sounds in a different order. Our research question is therefore:

How effectively can sequential modelling approaches recognise spoken Swahili words, and what evidence supports the strengths and limitations of each approach?

We answer it with five approaches that differ in *how* they model temporal order. A1 discards order: it is a logistic regression on MFCC statistics. A2 models order with a Markov chain: a GMM-HMM per word. A3 uses recurrence: a bidirectional LSTM with attention. A4 uses convolution over time: TC-ResNet. A5 uses self-attention after multilingual self-supervised pretraining: a fine-tuned XLS-R. All five share one frozen data split, one feature pipeline and one evaluation code base. They were tuned on validation only, in 55 logged development runs, and the test split was used only after every configuration was frozen.

Our contributions are:

- *A controlled comparison of five ways of modelling order* on an African-language speech task. Neural models are trained with three seeds, and the test results carry bootstrap confidence intervals and Holm-corrected McNemar tests (Section V).
- *Direct evidence of how the models use order*. A temporal-order stress test reverses or shuffles each word. It shows that all order-aware models depend on order, but that the pretrained A5 relies mainly on order within about 100 ms (Section VI).
- *A data-size curve*. With only 24 training clips per word, the pretrained model is already about as accurate as the models trained from scratch on all 245.
- *An error analysis without a human listener*. It combines confident learning with an external Swahili recogniser. The clips that defeat every order-aware model are far more often quiet, noisy or badly cropped, and include a likely label error and words said in English.
- *An open, reproducible pipeline*. The notebooks rebuild every table and figure in this report from the logged runs and predictions.

Section II reviews related work, Section III the data, and Section IV the methods and metrics. Section V reports and discusses the results, Section VI analyses the errors and limitations, and Section VII concludes.

II. RELATED WORK

A. From HMMs to neural keyword spotting

Recognising a word from a small vocabulary is one of the oldest speech tasks. Classic systems pair a compact spectral representation, the mel-frequency cepstrum, which outperformed other parametric representations for word recognition [5], with one hidden Markov model per word, trained by Baum–Welch re-estimation and decoded by maximum likelihood [6]. Our A1 and A2 baselines follow this tradition. A1 keeps only order-free statistics of the cepstral frames, and A2 is the per-word left-to-right HMM.

The task became a standard benchmark with the Speech Commands dataset, which also proposed a reproducible way to report keyword-spotting accuracy [7]. Convolutional networks then replaced HMM and DNN pipelines for small-footprint keyword spotting. Under fixed budgets on multiplications and parameters, they reduced the false-reject rate by 27–44% relative to a DNN [8]. Temporal convolutions over the frames, arranged in a compact residual network (TC-ResNet), made keyword spotting fast enough for real-time use on phones (a reported 385× speed-up) while surpassing the accuracy of the previous state of the art on Speech Commands [9]. This is the design behind our A4.

B. Recurrence and attention

Recurrent networks model a sequence by carrying a hidden state from frame to frame. Long short-term memory (LSTM) units keep the error signal constant through gated memory cells, and can learn dependencies across more than 1,000 time steps [10]. Stacked, bidirectional LSTMs brought this to acoustic modelling, reaching a 17.7% phoneme error rate on TIMIT [11]. Attention replaces a single summary vector with a learned, soft search over positions [12]. For speech commands, de Andrade et al. combined convolution, recurrence and attention in one model, reaching 94.1% and 94.5% accuracy on Speech Commands V1 and V2 with only 202K parameters. They also used the attention weights to show which parts of the audio drove each decision [13].

Our A3 belongs to this line. It differs in separating the contributions: A3 starts purely recurrent, adds the convolutional front-end only as a tested search step, and is compared with the purely convolutional A4.

Whether recurrence is needed at all is itself an open question. A systematic comparison across sequence-modelling benchmarks found that a simple convolutional architecture outperforms canonical recurrent networks such as LSTMs, while also showing a longer effective memory [14]. Our A3–A4 comparison tests whether this holds for sequences as short as a single spoken word, with only a few hundred training examples per class.

C. Augmentation for small speech datasets

SpecAugment showed that simple transformations applied directly to filter-bank features (time warping, and masking blocks of frequency channels and of time steps) are enough to reach state-of-the-art results on large speech-recognition

benchmarks [15]. Our setting is the opposite extreme: about 245 training clips per word, and competition rules that forbid external data such as noise corpora. Synthetic, feature-level augmentation is therefore the only augmentation available to us, and its value is tested directly in round R2.

D. Self-supervised speech models and African languages

Transformers relate every position of a sequence to every other through attention, without recurrence or convolution [16]. For keyword spotting, a Transformer trained from scratch matched the state of the art on Speech Commands, with 98.6% and 97.7% accuracy on the 12- and 35-word tasks [17]. The Audio Spectrogram Transformer reached 98.1% [18]. Both used far more labelled audio than we have, and the second also depended on ImageNet pretraining, because “the Transformer needs more data to train” [18].

Self-supervised pretraining removes most of the need for labels. wav2vec 2.0 learns from unlabelled audio by masking spans of latent speech frames and identifying the true frames from context. Fine-tuned on only 10 minutes of transcribed speech, it reached 4.8% and 8.2% word error rate on the two LibriSpeech test sets [19]. Pretraining on many languages at once helps low-resource languages further [20]. XLS-R scaled this to 436K hours in 128 languages [21], and MMS to 1,406 languages [22]. The SUPERB benchmark showed that such models reach about 96% keyword-spotting accuracy on English Speech Commands even when frozen, and that a weighted sum of their hidden layers is often a better input than the last layer alone [23].

For African languages, labelled speech remains scarce. Swahili belongs to the “Hopefuls”, a class of languages with only limited labelled resources [2]. It was absent from the first releases of the crowdsourced Common Voice corpus [3], and it makes up only 91 of XLS-R’s 436K pretraining hours [21]. Speech technology for such languages has therefore turned to keyword spotting without full speech recognition, for example to monitor radio for humanitarian programmes [24], and to self-supervised pretraining on unlabelled radio archives [1]. Our study adds a controlled comparison for an African language: how much a pretrained Transformer adds over classical and from-scratch sequence models, under the same data, split and metrics.

E. How the literature shaped our study

Four lessons from this work shaped the design:

- 1) Keyword spotting has been solved with very different ways of modelling time: order-free features, Markov states, convolution, recurrence and self-attention. We therefore compare one representative of each on the same data, split and metrics, rather than tuning a single family.
- 2) Attention-based recognisers can show where in the audio they look [13]. We inspect A3’s attention, but test our central hypothesis through behaviour: words whose

sounds differ mainly in order (*tisa* and *sita*) should separate order-aware from order-free models, and reversing a word should hurt only the former.

- 3) Feature-level augmentation such as SpecAugment improves recognition even on large corpora [15]. With only 245 clips per word, we treat it as a first-class experimental factor rather than a fixed default.
- 4) Pretrained speech Transformers need very little labelled data [19], but their pretraining data contain little Swahili [21]. A5 therefore tests transfer from other languages, and its R2 search compares multilingual with English-only pretraining.

III. DATASET AND EXPLORATORY ANALYSIS

A. Data

We use the Swahili Audio Classification dataset [4]. It holds recordings of 12 spoken Swahili words, made by about 300 crowd-workers in Kenya: the digits *moja* to *kumi* (one to ten), *ndio* (yes) and *hapana* (no). There are 4,200 labelled clips, exactly 350 per word, and 1,800 unlabelled competition test clips. All 6,000 files are 16 kHz mono 16-bit PCM, with no corrupt files and no byte-identical duplicates. The competition allows no external data and forbids redistribution, so the audio is kept off the public repository. As described in Section IV, we froze one stratified 70/15/15 split (2,940 / 630 / 630 clips). The distribution statistics below use the training split only. Analyses that score a model (the confusions, the prefix curve and the normalisation check) use the validation split, as model selection does. None uses the test split.

B. What the data revealed

a) The words are short, but the clips are long: Clips last 2.5–62 s (median 4.3 s), while the spoken word lasts a median 0.55 s. Word duration ranges from 0.45 s for *tatu* to 0.65 s for the three-syllable *hapana*, and 89.5% of words last at most 1.0 s (Fig. 1). A 30 dB silence trim leaves a median of 1.9 s and a 95th percentile of 5.5 s, because background noise is kept as part of the “word”. The models therefore read a centred window around the clip’s highest-energy region: 1.5 s by default, 1.0 s for the final A3 and 2.0 s for the final A4. The 1.5 s window retains a median 99.8% of each clip’s above-noise energy.

b) Recordings vary in level, but are mostly clean: Levels span about 19 dB between the 5th and 95th percentiles (median −32 dBFS), and 3.8% of clips reach full scale, so windows are peak-normalised. The median estimated SNR is 42 dB, and 80% of clips exceed 30 dB. Our synthetic noise augmentation (10–30 dB) is therefore harsher than most real recordings. The noisy minority matters, however: the models fail most on clips below 30 dB (Section VI).

c) The vocabulary contains an anagram: *sita* (six) and *tisa* (nine) contain the same sounds in a different order. The average spectrograms show it directly (Fig. 2): the high-frequency energy of /s/ comes at the start of *sita* and in the middle of *tisa*. Other near-pairs are *nne/nane* (letter edit distance 1), and *tano/tatu* and *saba/sita* (distance 2). We track

TABLE I
HOW THE EXPLORATORY ANALYSIS SHAPED THE DESIGN.

Finding	Decision
350 clips per word, perfectly balanced	No class weighting; accuracy, macro-F1 and log loss
Words last a median 0.55 s in 2.5–62 s clips; silence trimming leaves noisy tails	Centred 1.5 s energy window; 1.0/2.0 s tested
Levels vary by ~19 dB; median SNR 42 dB	Peak normalisation; CMVN for the HMM
<i>sitaltisa</i> differ only in where /s/ occurs	Sequence models; four sound-alike pairs tracked
Order-free features do not cluster by word	Per-utterance normalisation; order-aware models
Evidence arrives through the word	Models see the whole word
No near-duplicates; no speaker IDs	No leakage found; speaker overlap is a limitation

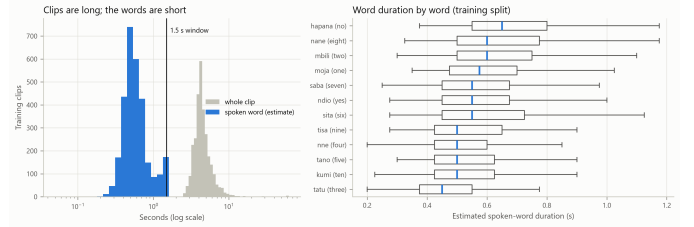


Fig. 1. Left: whole clips are far longer than the spoken words they contain (training split, log scale). Right: estimated word duration per word.

these four pairs for every model. Spelling similarity, however, only weakly predicts what an order-free model confuses (Spearman $\rho = -0.18$ over all 66 pairs, $p = 0.16$). Its most confused pair on validation, *mbili/ndio*, is far apart in spelling, but both words begin with a prenasalised stop and contain /i/.

d) Order-free features do not separate the words: A t-SNE map of order-free MFCC statistics shows no word forming its own cluster. The only distinct islands contain clips of many words, so they probably group recordings by speaker or recording condition. This is consistent with the order-free baseline’s ceiling of 77.9%, and with the large gain from per-utterance normalisation in the HMM (53% to 76% accuracy).

e) The evidence builds up through the word: Trained on only the first 25/50/75/100% of each word, the order-free classifier reaches 49/72/80/80% accuracy. Words that share a first syllable (*tano*, *tatu*) separate only once their second syllable begins. The evidence therefore builds up over most of the word, not just its onset, and the window must cover the whole word.

f) No leakage, but speakers are unknown: No pair of clips across the splits is a near-duplicate (the maximum cosine similarity of standardised MFCC statistics is 0.94). Median pitch spans 106–235 Hz between the 10th and 90th percentiles, consistent with a mix of adult male and female voices. Speaker identities are not provided, however, so the same speaker may appear in both training and test clips (Section VI-B).

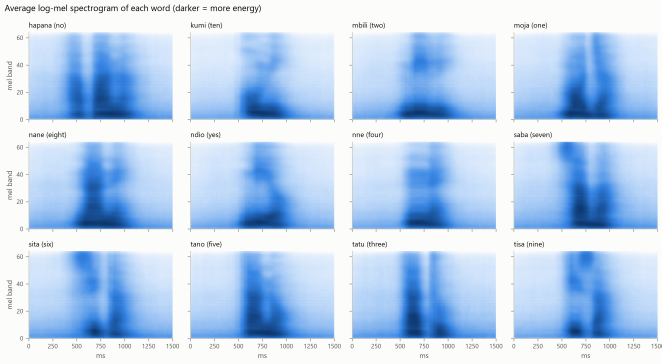


Fig. 2. Average log-mel spectrogram of each word in the centred 1.5 s window. The high-frequency /s/ comes at the start of *sita* and in the middle of *tisa*.

IV. METHODOLOGY

A. Data and experimental protocol

The challenge training set contains 4,200 labelled clips (350 per word), all 16 kHz mono 16-bit PCM [4]. The test labels are hidden, so we froze one stratified 70/15/15 split of the training set (seed 42) into 2,940 training, 630 validation and 630 test clips (52–53 per word in validation and test). A hash of the split is checked by every notebook. No byte-identical duplicate recordings were found. Model selection uses the validation split only. The test split is used only after every configuration is frozen, for the final runs and the data-size runs, and never for selection. Experiments proceed in rounds: R0 establishes each approach, R1 trains default neural configurations, R2 tunes, R3 trains the final configurations with three seeds and evaluates them on the test split, and R4 varies the amount of training data. Every run is logged with the change it makes and the reason for it (`results/experiments.csv`).

B. Preprocessing

Clips last 2.5–62 s (median 4.3 s), but a spoken digit lasts under a second, so most of each clip is silence or background noise. A single silence-trimming threshold was unreliable: at 30 dB, 10% of trimmed clips in a 600-clip sample still exceeded 4.5 s. We therefore keep the window with the highest signal energy: 1.5 s by default, with 1.0 and 2.0 s tested in R2. When the whole word fits, many window positions hold almost the same energy, so we take the middle of that plateau. This centres the word: the energy centroid lies within ± 0.2 s of the window centre for 90% of clips. The 1.5 s window retains a median of 99.8% of each clip’s above-noise energy, and less than 70% in only 2.7% of clips, which are mostly long recordings containing several utterances. Each window is peak-normalised.

C. Acoustic features

Frames use a 25 ms window and a 10 ms hop, giving 151 frames per window. We compute a 64-band log-mel spectrogram (dB, 80 dB dynamic range) and mel-frequency cepstral coefficients (MFCCs) [5] with delta and delta-delta coefficients (a 9-frame window). Features are computed once,

cached, and shared by all members, so every approach sees identical inputs. Normalisation statistics are always fitted on the training split only.

D. Approaches

Table II summarises the five approaches. They were chosen to vary *how* temporal order is modelled: not at all (A1), by a generative Markov chain (A2), by recurrence (A3), by convolution over time (A4) and by self-attention with self-supervised pretraining (A5).

a) *A1: MFCC statistics with a linear classifier*: Each clip becomes one 156-dimensional vector: the mean, standard deviation, minimum and maximum over time of 13 MFCCs with their deltas. This discards temporal order, except for the roughly 90 ms of local change that the deltas span. After standardisation, a multinomial logistic regression ($C = 0.1$, chosen by 5-fold cross-validation on the training split) outperformed an RBF-kernel SVM with Platt-scaled probabilities [25], [26].

b) *A2: GMM-HMM per word*: Following the classic isolated-word recogniser [6], each word has its own left-to-right HMM: each state either repeats or moves to the next. Each state emits diagonal-covariance Gaussian mixtures over 39-dimensional frames (13 MFCCs with deltas). The frames are normalised per utterance (cepstral mean and variance normalisation, CMVN) to remove the channel differences of about 300 speakers recording on different phones. States are initialised by a flat start: every training sequence is split uniformly into segments, and state i is initialised from the i -th segments. Training uses Baum–Welch, with small MAP priors on the mixture weights, variances and allowed transitions. These priors prevent components starved of frames from producing degenerate (NaN) parameters. A clip is assigned to the word whose model gives the highest per-frame log-likelihood. The final model has 12 states and 8 Gaussians per state.

c) *A3: bidirectional LSTM with attention*: The log-mel frames (151×64) feed two stacked bidirectional LSTM layers [10], [11] with 128 units per direction. The outputs are pooled over time with additive attention [12], followed by dropout and a linear layer over the 12 words. Recurrence carries information across the whole word, and attention learns which frames matter. It also exposes, for the error analysis, which frames each decision relied on. The design follows the attention-based recogniser of de Andrade et al. [13] with one deliberate difference. We omit their convolutional layers by default, so that A3 is purely recurrent and contrasts cleanly with the convolutional A4. Adding them back is one of the R2 ablations.

The R2 search selected 40 MFCCs with deltas instead of log-mel, a 1.0 s window, SpecAugment, a peak learning rate of 2×10^{-3} , and de Andrade et al.’s convolutional front-end (two 1-D convolutions, kernel 5, 128 channels). The final model has 855K parameters. The front-end lowered the validation log loss only marginally ($0.142 \rightarrow 0.134$) and did not change the accuracy (97.0%), so the recurrence does most of the work.

The comparisons with A4 use this final model; the stress test also includes the purely recurrent variant (687K parameters).

d) *A4: temporal convolutional network (TC-ResNet)*: A4 follows TC-ResNet [9]. It treats the frequency axis as channels, and slides one-dimensional convolutions over time only. A three-wide stem convolution feeds residual blocks. Each block has two convolutions of kernel width 9, the first with stride 2, and a projected shortcut when the shape changes. Global average pooling over time feeds a linear layer over the 12 words.

Our default is TC-ResNet8 at width 1.5: three blocks with 24–72 channels. It has 146K parameters, and a receptive field of 171 frames (1.7 s), which already spans the whole input window. Temporal order therefore reaches A4 only through the arrangement of its strided, stacked filters. There is no state carried through time, as there is in A3. Because the classifier follows global average pooling, applying it at every time step gives an exact class activation map over time, which we use for interpretation. The R2 search (below) kept this architecture, and selected MFCC-40 with Δ and $\Delta\Delta$ (120 channels) from a 2.0 s window, with full augmentation. The final model has 150K parameters (configs/a4_best.yaml), 5.7 times fewer than the final A3.

e) *A5: fine-tuned self-supervised speech Transformer*: A5 fine-tunes a Transformer [16] speech encoder that was pretrained without labels, following wav2vec 2.0 [19]. A convolutional feature encoder turns the raw 16 kHz waveform of the same centred window into one vector per 20 ms. A Transformer then contextualises these vectors, relating every frame to every other through self-attention. Order enters only through a convolutional positional embedding. The frames of the last layer are projected to 256 dimensions, mean-pooled over time, and classified.

The default backbone is XLS-R [21] with 300M parameters, pretrained on 436K hours in 128 languages, including 91 hours of Swahili. As is standard, the convolutional feature encoder stays frozen, and the Transformer is fine-tuned. The new, randomly initialised head has its own, higher learning rate. The only built-in regularisation is wav2vec 2.0’s masking of latent frames: spans of 10 frames, started with probability 0.05. Both backbones in the R2 comparison use these settings, although they were pretrained with different defaults. The R2 search kept this setup and added waveform augmentation. The final model has 316M parameters, of which 311.5M are fine-tuned (configs/a5_best.yaml): 370 times more than A3 and 2,100 times more than A4.

E. Training the neural models

A3 and A4 share one training loop:

- *Optimisation*: AdamW with a one-cycle learning-rate schedule, gradient clipping at 1.0, cross-entropy with label smoothing 0.1, batch size 64, and at most 40 epochs.
- *Early stopping*: training stops when the validation log loss (without label smoothing) has not improved for five epochs, and the weights of the best epoch are restored.

TABLE II
THE FIVE APPROACHES AND HOW EACH TREATS TEMPORAL ORDER.

ID	Model	Temporal order
A1	MFCC statistics + logistic regression	Discarded (bag of frames); Δ features keep ~90 ms of local change
A2	GMM-HMM per word	Left-to-right Markov states
A3	BiLSTM + attention	Recurrence in both directions
A4	TC-ResNet	Convolution over time
A5	XLS-R (fine-tuned)	Self-attention; multilingual self-supervised pretraining

- *Augmentation*: applied on the fly to the feature frames of training clips, using SpecAugment time and frequency masking [15], time shifts of up to 100 ms and tempo perturbation (0.9–1.1 $\times$). Noise injection at 10–30 dB SNR assumes a dB power scale, so it is used for log-mel inputs only. Padding and masks take the clip’s minimum and mean value. For log-mel, whose bands share one dB scale, that is one value per clip. A4’s MFCC inputs use one value per channel, because their channels differ in scale by orders of magnitude. With a single value, the first MFCC run of A4 collapsed to chance. All augmentation is synthetic, because the competition allows no external data.
- *Reproducibility*: every generator is seeded before the weights are initialised, so a configuration reproduces exactly.

A5 uses the same loop, instead of a separate fine-tuning framework, so that the loss, model selection and calibration are identical across the neural models. Four settings differ, as fine-tuning a pretrained model requires:

- each waveform is normalised to zero mean and unit variance inside the model, instead of standardised with training-split statistics;
- the pretrained Transformer is fine-tuned at a peak learning rate of 3×10^{-5} , and the new head at 10^{-3} , with 10% warm-up;
- training runs for 10 epochs with batch 16 and keeps the best epoch, with no early stopping;
- augmentation acts on the waveform: time shifts, speed perturbation (0.9–1.1 $\times$, which unlike the log-mel tempo change also shifts the pitch), and noise at 10–30 dB SNR.

The probabilities of A2–A5 are then temperature-scaled [27], as described in Section IV-H. A1’s fitted temperature was 0.99, so it is used unscaled.

F. Hyperparameter search

Classical models were tuned by grid search: cross-validated on the training split for A1, and on the validation split for the HMM topology and preprocessing of A2. For A3 we ran a greedy, one-factor-at-a-time search in round R2. Starting from the default configuration, each run changes a single factor relative to the current best configuration, and keeps the change only if the validation log loss improves. The factors were

augmentation, pooling (attention, last state, mean), recurrent direction, input features (log-mel or MFCC), window length (1.0/1.5/2.0 s), the way the word is located, the learning rate, and a convolutional front-end. A4 followed the same protocol over the factors that apply to it: augmentation, input features, window length, the way the word is located, depth, kernel width, network width and the learning rate. A5’s search covered the pretraining language (XLS-R against the English-only wav2vec 2.0 Base [19]), freezing the whole encoder, augmentation, the window length, a weighted sum of all layers instead of the last [23], and the learning rate. This makes every decision traceable to a logged comparison. The cost is that interactions between factors are only explored along the chosen path. Runs that failed, or that a later fix made stale, are kept outside the results table (`results/failed_runs/` and `results/superseded_runs/`).

Single-seed comparisons are noisy, so we measured the noise. Retraining the final A3 configuration with three seeds (42, 13, 7) gave a validation log loss of 0.137 ± 0.004 (mean $\pm$ SD) and an accuracy of $96.9 \pm 0.2\%$. Training is deterministic: the seed-42 run reproduced the R2 result exactly. Search steps that changed the log loss by about one SD are therefore within seed noise. These include the window length (1.0 versus 1.5 s), the learning rate and the rejection of the 2.0 s window. We report them, but draw no conclusions from them. Only the larger effects, such as augmentation, pooling, direction and the way the word is located, are interpreted.

A4 is much less stable across seeds: 0.135 ± 0.023 log loss and $96.8 \pm 0.3\%$ accuracy. One seed stopped early, at epoch 16 of 40, while the learning rate was still high. For A4, therefore, only the effect of the way the word is located (silence trim: $+0.157$ log loss) is clearly larger than seed noise. The other R2 differences (at most 0.039) are read as directions, not established effects. A5 is the most stable model: 0.1015 ± 0.0009 log loss and $98.0 \pm 0.1\%$ accuracy. Pretrained weights leave the seed little to decide. So all of A5’s R2 differences, except the tie between the 1.5 and 2.0 s windows, are larger than its seed noise.

G. Implementation

Everything is written in Python. Features use librosa [28]; A1 uses scikit-learn [29], A2 hmmlearn [30], A3–A5 PyTorch [31], and A5 the Hugging Face Transformers library [32]. Neural models were trained on a single NVIDIA Tesla T4 (Kaggle), except one A4 run on a laptop CPU, and the classical models on CPUs. The repository’s notebooks rebuild every table and figure in this report from the logged runs and predictions (`results/`).

H. Evaluation Metrics

The task is closed-set classification of a clip into one of $K = 12$ words. Every model outputs a probability vector $\hat{\mathbf{p}}_i$, and every model is scored by the same code (`src/evaluate.py`), so the numbers are comparable across approaches. Table III summarises the metrics and what each adds.

a) *Log loss (co-primary)*: The official challenge metric [4] is the multi-class log loss,

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \hat{p}_{i,y_i}, \quad (1)$$

where y_i is the true word of clip i . Log loss is a strictly proper scoring rule: its expected value is minimised only by reporting the true class probabilities, so it rewards honest confidence rather than just the top-ranked label [33]. This matters for the intended use. A voice interface for mobile money or a helpline must decide whether to act on a recognised digit or ask the caller to repeat it, and that decision depends on the confidence. Log loss also punishes confident mistakes severely. Assigning probability 0.01 to the true word costs 4.6 nats, against $\ln 12 \approx 2.48$ for a uniform guess.

Our own experiments show why accuracy alone is not enough. For the MFCC-statistics SVM (an early A1 variant on 40 MFCCs), the probability method changed the log loss far more than the accuracy. libsvm’s pairwise-coupled Platt probabilities [25], [26] gave a validation log loss of 0.880 at 74.3% accuracy. The one-vs-rest sigmoid calibration that scikit-learn now recommends gave 1.230 at 73.5%, and temperature calibration gave 1.153 at 73.7%. Accuracy is almost unchanged, yet log loss is 40% worse.

b) *Accuracy and macro-F1 (co-primary)*: The training set is exactly balanced (350 clips per word) and the frozen validation and test splits hold 52–53 clips per word. Accuracy is therefore not inflated by a majority class, and it is the conventional headline for keyword spotting [7]. We also report macro-F1, the unweighted mean of per-class F1 [34]. It penalises a model that over-predicts one word, which accuracy on a balanced set only partly reveals. We compute it as the mean of per-class F1 scores, not the F1 of mean precision and recall, since the two definitions can diverge [35]. Per-class precision, recall and F1, and the 12×12 confusion matrix, show which words fail. For example, A1’s recall is only 66% for *mbili*, which it confuses with *ndio*.

c) *Calibration*: We measure calibration with the expected calibration error (ECE, 15 equal-width confidence bins) [27], [36] and with reliability diagrams. Poor calibration is corrected post hoc with temperature scaling [27]: a single scalar T divides the logits (for the HMM, the per-frame log-likelihoods) before the softmax. It changes the confidence but never the predicted class, so accuracy and F1 are unaffected. To keep the validation log loss honest, T is *cross-fitted*: each validation fold is calibrated with a T fitted on the other four folds. T is then refitted on the whole validation split for use on the test split.

d) *Task-specific diagnostic: sound-alike pairs*: The research question is whether modelling temporal order helps. We therefore track, for word pairs that share sounds, the share of the pair’s clips predicted as the other word of the pair. The pairs are *tisalsita* (an anagram: the same sounds in a different order), *nnelnane*, *tatultano* and *sabalsita*. This links the evaluation directly to the hypothesis. The order-free A1

TABLE III
EVALUATION METRICS AND THEIR ROLE.

Metric	Why we use it	Limitation
Log loss	Official metric; proper scoring rule; rewards honest confidence	Dominated by a few confident errors
Accuracy	Balanced classes; standard for keyword spotting	Ignores confidence
Macro-F1, per class	Exposes words a model systematically fails	Not a probability metric
ECE + reliability	Directly measures calibration	Depends on binning; sparse bins
Pair confusion	Tests the temporal-order hypothesis	About 105 clips per pair
Bootstrap CI, McNemar	Separates real differences from noise	One split; no speaker grouping

confuses *tisalsita* on 6.7% of their validation clips, while the order-aware HMM (A2) confuses them on none.

e) Uncertainty and significance: Model selection uses the validation split only. The test split is used only after every configuration is frozen, and never for selection. Neural models are trained with three seeds and reported as mean $\pm$ standard deviation. For every model we compute 95% percentile bootstrap confidence intervals for accuracy and log loss (1,000 resamples) [37]; they are listed in `results/metrics/test_results.csv`. Pairs of models are compared on the same test clips with the exact McNemar test on their discordant predictions [38]. Dietterich found that this test has an acceptable Type I error, and recommends it when each learning algorithm can be run only once [39]. That fits our setting: retraining the pretrained Transformer the ten times a 5×2 cross-validated test requires is too expensive. The five planned comparisons (the best model against each other model, and the recurrent A3 against the convolutional A4) are corrected together for multiplicity with Holm’s procedure [40].

f) Limitations: Selection bias, the size of the test split, unknown speakers and the closed vocabulary limit what these metrics can show. They are discussed in Section VI-B.

V. RESULTS AND DISCUSSION

A. How the configurations were reached

Table IV lists the development decisions with the largest effects. It covers 55 logged runs, all scored on validation.

Two decisions had the largest effects. The first is locating the word. Replacing the centred energy window with silence trimming raised the validation log loss of A3 from 0.144 to 0.759 and of A4 from 0.119 to 0.276. For the HMM, per-utterance CMVN and the energy window together cut the log loss from 1.65 to 0.38, before any change to the model. The second, tested on A3, is how the sequence is summarised. A3 with its last hidden state instead of attention fell from 96.0% to 69.7% accuracy, and a forward-only A3 lost 0.09 log loss.

Augmentation helped every neural model, but in different forms. SpecAugment gave A3 its largest tuning gain (0.214 to

0.153), but adding time shift, tempo and noise on top made it worse (0.166). For A4, SpecAugment alone changed nothing (0.176 to 0.179), while the full set (SpecAugment, time shift, tempo and noise) improved it to 0.140, a gain of about 1.5 of its seed standard deviations. The waveform counterpart improved A5 (0.111 to 0.101).

Pretraining needs fine-tuning. A5 with a frozen encoder reached only 27% accuracy. English-only wav2vec 2.0 Base came close to the multilingual XLS-R (0.120 against 0.111), although it is 3.3 times smaller.

Some differences are within seed noise. Retraining each final configuration with three seeds gave standard deviations of 0.004 (A3), 0.023 (A4) and 0.001 (A5) in validation log loss. Differences smaller than these are not interpreted.

B. Test results

Table V gives the final scores on the 630 test clips. A5 is the best model on every metric: log loss 0.093 ± 0.001 , accuracy $98.1 \pm 0.1\%$ and macro-F1 0.982 (12 errors for seed 42). A4 follows (log loss 0.125, 96.3%), then A3 (0.170, 96.0%), the HMM (0.186, 96.5%) and the order-free A1 (0.700, 79.5%).

Validation predicted test well. A5, A2 and A1 moved by at most 0.025 log loss, and A4 improved slightly. A3 lost the most (0.137 to 0.170). Its configuration combines five search decisions, each chosen on validation, so this is consistent with selection optimism [41]. With 630 clips, the seed-42 bootstrap intervals span about ± 1 –1.5 accuracy points for the order-aware models (for A5, 97.0–99.0%) and ± 3 for A1.

a) Which differences are real: A5 is significantly more accurate than every other model after Holm’s correction (Table VI). On log loss, its lead over A1–A3 is clear, but its lead over A4 is not: the interval includes zero. A5 makes fewer errors, while A4’s probabilities are nearly as good. The central architectural comparison, recurrence (A3) against convolution (A4), shows no significant difference in accuracy (4 against 9 discordant clips, $p = 0.27$). A4’s log loss is lower, however, by 0.059 [0.018, 0.101]. That holds for seed 42; across seeds the gap (0.170 against 0.125) is under two of A4’s seed standard deviations. The final A3 also has a convolutional front-end, so this compares a recurrent network with convolutional input layers against a purely convolutional one. The HMM ties with both on accuracy as well (11 against 9 discordant clips with A3, $p = 0.82$; 5 against 8 with A4, $p = 0.58$; uncorrected tests outside the family of Table VI).

b) Calibration: Label smoothing leaves every neural model under-confident, and the HMM’s raw scores are flatter still (test ECE 0.07–0.23 before scaling). Temperatures fitted on validation (0.38–0.71) transfer to the test split and bring ECE down to 0.011–0.023.

c) Where the errors are: The anagram pair shows the effect of modelling order: *sitaltisa* is confused in 8.6% of its test clips by A1, but in only 1.0–1.9% by A2–A5. Per word, A1 fails broadly, while the order-aware models’ errors are spread thinly over many words (Fig. 3). Section VI shows that ten recordings, shared by all order-aware models, account for 10 of A5’s 12 errors and about half of the other models’ errors.

TABLE IV

DEVELOPMENT DECISIONS WITH THE LARGEST EFFECTS (VALIDATION LOG LOSS, SEED 42). EACH ROW CHANGES THE THEN-BEST CONFIGURATION; THE A2 GRID AND A4'S FULL AUGMENTATION CHANGE SEVERAL SETTINGS AT ONCE. THE SEED STANDARD DEVIATION OF THE FINAL CONFIGURATIONS IS 0.004 (A3), 0.023 (A4) AND 0.001 (A5).

Approach	Change	Before	After	Decision
A1	RBF SVM $\rightarrow$ multinomial logistic regression	0.783	0.725	kept (final)
A1	13 $\rightarrow$ 40 MFCCs in the statistics	0.783	0.880	rejected
A2	+ per-utterance CMVN	1.650	0.931	kept
A2	silence trim $\rightarrow$ centred 1.5 s energy window	0.931	0.380	kept
A2	5 states $\times$ 2 Gaussians $\rightarrow$ 12 $\times$ 8 (grid search)	0.380	0.180	kept (final)
A3	+ SpecAugment	0.214	0.153	kept
A3	SpecAugment $\rightarrow$ full augmentation	0.153	0.166	rejected
A3	attention pooling $\rightarrow$ last hidden state	0.153	0.915	rejected
A3	bidirectional $\rightarrow$ forward-only LSTM	0.153	0.246	rejected
A3	energy window $\rightarrow$ silence trim	0.144	0.759	rejected
A3	+ convolutional front-end	0.142	0.134	kept (final)
A4	+ SpecAugment	0.176	0.179	rejected
A4	+ full augmentation (SpecAugment, shift, tempo, noise)	0.176	0.140	kept
A4	log-mel $\rightarrow$ MFCC-40 with deltas (noise then off)	0.140	0.133	kept
A4	1.5 s $\rightarrow$ 2.0 s window	0.133	0.119	kept (final)
A4	energy window $\rightarrow$ silence trim	0.119	0.276	rejected
A4	deeper / wider kernels / narrower network	0.119	0.151–0.158	rejected
A5	multilingual XLS-R $\rightarrow$ English-only wav2vec 2.0 Base	0.111	0.120	rejected
A5	fine-tuned $\rightarrow$ frozen encoder (head only)	0.111	1.960	rejected
A5	+ waveform augmentation (shift, speed, noise)	0.111	0.101	kept (final)
A5	1.5 s $\rightarrow$ 1.0 s window	0.101	0.130	rejected

TABLE V

TEST RESULTS (630 CLIPS; MEAN $\pm$ SD OVER THREE SEEDS FOR A3–A5). PARAMS COUNTS ALL WEIGHTS. TRAIN IS THE FINAL RUN'S TRAINING TIME (T4 GPU FOR A3–A5, CPU FOR A1–A2). CPU MS/CLIP IS THE LATENCY FROM AUDIO FILE TO PROBABILITIES ON ONE CPU THREAD.

Approach	Params	Train (min)	CPU ms/clip	Log loss	Accuracy (%)	Macro-F1	ECE
A1 MFCC stats + LR	2K	0.1	7	0.700	79.5	0.795	0.047
A2 GMM-HMM	93K	11.0	113	0.186	96.5	0.965	0.023
A3 BiLSTM + attention	855K	1.2	14	0.170 $\pm$ 0.009	96.0 $\pm$ 0.3	0.960 $\pm$ 0.003	0.017 $\pm$ 0.001
A4 TC-ResNet	150K	1.0	9	0.125 $\pm$ 0.026	96.3 $\pm$ 1.2	0.963 $\pm$ 0.012	0.019 $\pm$ 0.003
A5 XLS-R (fine-tuned)	316M	13.0	871	0.093 $\pm$ 0.001	98.1 $\pm$ 0.1	0.982 $\pm$ 0.001	0.014 $\pm$ 0.003

TABLE VI

SIGNIFICANCE ON THE TEST SPLIT (SEED 42). McNemar's EXACT TEST COUNTS THE CLIPS WHERE ONLY ONE MODEL IS RIGHT; p IS HOLM-CORRECTED OVER THE FIVE COMPARISONS. Δ IS THE DIFFERENCE IN LOG LOSS (FIRST – SECOND) WITH A PAIRED-BOOTSTRAP 95% INTERVAL.

Comparison	Only 1st	Only 2nd	p	Δ log loss [95% CI]
A5 vs A1	118	1	$< 10^{-32}$	-0.608 [-0.698, -0.531]
A5 vs A2	12	2	0.039	-0.094 [-0.133, -0.055]
A5 vs A3	13	1	0.007	-0.074 [-0.121, -0.029]
A5 vs A4	7	0	0.039	-0.014 [-0.036, +0.005]
A3 vs A4	4	9	0.27	+0.059 [+0.018, +0.101]

C. How much labelled speech is needed

Each final configuration was retrained on 10, 25 and 50% of the training clips (nested, stratified subsets; one seed each) and scored on the test split (Fig. 4).

- *Pretraining pays off most when data is scarce.* With 24 clips per word, A5 reaches 96.5%. That is about as accurate as the models trained from scratch on all 245 clips per word (96.2–97.0%). It saturates by about 120

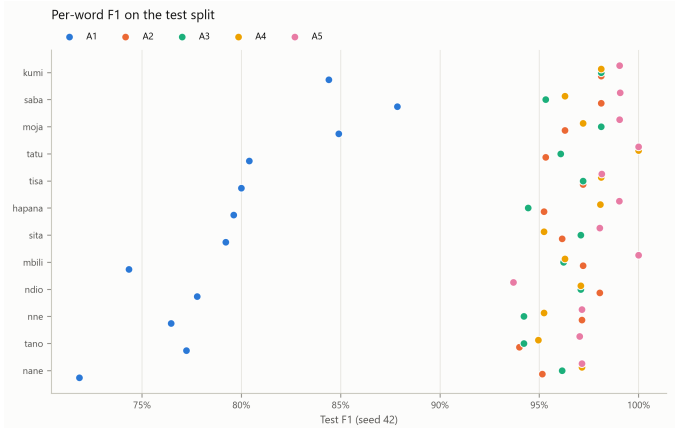


Fig. 3. Per-word test F1 (seed 42). The order-free A1 is far behind on every word. The order-aware models stay above 93% on every word.

clips per word.

- *Without pretraining, the HMM is the most data-efficient*

model. At 24 clips per word it reaches 89.2%, against 83.5% for A3 and 62.5% for A4. From 61 clips per word on, A3 and A4 stay within 1.4 points of the HMM, and only on the full training set does A4 overtake it.

The recipe was held fixed, so the smallest subsets also received fewer optimisation steps. At 10%, A3–A5 were still improving at their final epoch, so those points understate them, A4 most of all.

D. Cost

The accuracy has a price (Table V). A5 has 316M parameters (1.26 GB as 32-bit weights) and needs 871 ms per clip on one CPU thread. A4 has 150K parameters (0.6 MB) and needs 9 ms. A5 is thus about 96 times slower and 2,100 times larger, in exchange for 0.032 lower log loss and 1.8 more accuracy points. On a phone or a low-cost voice server, A4 is the efficient choice. Where an error is costly, for example when confirming a mobile-money amount, A5 is worth its price.

E. Discussion

We return to the research question: how effectively do sequential models recognise spoken Swahili words, and why?

- 1) *Modelling order is necessary*. Every order-aware model reaches 96–98% test accuracy, against 79.5% for order-free statistics. A1 also differs in features and capacity, so the gap alone does not prove that order is the cause. The anagram pair and the stress test (Section VI) make the link directly, as the EDA predicted: the order-aware models confuse *sita* and *tisa* rarely, and fail on reversed words. For A5, the order that matters lies mostly within about 100 ms.
- 2) *How order is modelled matters less than expected*. A classical HMM ties the neural models trained from scratch on accuracy. Convolution matches a BiLSTM with a convolutional front-end in accuracy, with 5.7 times fewer parameters, and has a lower log loss for seed 42. This agrees with the finding that convolutional networks rival recurrent ones on sequence tasks [14], here for sequences as short as a single word.
- 3) *Pretraining brings the largest gain we observed*, above all with little data. A model that heard only 91 hours of Swahili during pretraining [21], and an English-only model almost as well, transfer effectively to Swahili words. Our design confounds pretraining with A5’s architecture, input and size (Section VI-B), but the result supports self-supervised pretraining as the practical route for low-resource languages [1], [19].
- 4) *The data sets the ceiling*. Locating the word in a long, noisy recording mattered as much as the model. Ten test clips defeat every order-aware architecture. They are quiet, noisy or badly cropped far more often than other clips, and include a likely label error and words said in English (Section VI).

VI. ERROR ANALYSIS AND LIMITATIONS

A. Error analysis

The analysis uses the test split and the final seed-42 runs, unless stated otherwise. Confusion rates for A3–A5 are pooled over their three seeds.

a) *The order-aware models use the order of the sounds*: On the test split, the anagram pair *sita/tisa* is the order-free A1’s most-confused pair: 8.6% of the pair’s clips are heard as the other word. For A2–A5 the rate falls to 1.0–1.9%. Beyond that pair, spelling similarity does not predict confusion. Across all 66 word pairs, the rank correlation between confusion rate and edit distance lies between -0.14 and $+0.04$ for every model (all $p > 0.2$). The closest pair, *nane/nne* (edit distance 1), is confused in 4.7% of its clips by A1, but in at most 0.3% by the order-aware models.

The temporal-order stress test (Fig. 5) checks the use of order directly. Each validation clip is played unchanged, reversed, or cut into 100 ms chunks in random order. Every version keeps the same overall spectral content, although reversal also reverses the dynamics within each sound, such as a stop’s burst.

- *Reversal*: accuracy falls from 95–98% to 21–39% for A2–A5. On reversed audio, *tisa* is mostly heard as *sita* and vice versa: in 83% of the pair’s clips for the HMM and 62% for A5.
- *Control*: a logistic regression on static MFCC statistics alone is exactly unchanged by reversal (57.5%).
- *A1*: its delta features encode about 90 ms of local change, so it loses 28 points. It is order-free only beyond that scale.
- *Shuffling*: A5 keeps 88.9% accuracy when the chunks are shuffled, while A2–A4 fall to 32–61%. So the pretrained model relies on the order *within* chunks of about 100 ms (the transitions between sounds), and hardly on how those pieces are arranged. For the MFCC-based models the comparison is less clean, because the cuts also distort their delta features.

b) *The models fail on the same clips*: 493 of the 630 test clips (78%) are recognised by all five models, and 102 defeat only A1. Errors that belong to a single order-aware architecture are rare: A5 makes none that A2–A4 all avoid, A4 one, A2 seven and A3 eight. The same 10 test clips defeat A2, A3, A4 and A5 alike. They make up 10 of A5’s 12 errors, and 10 of the 19–24 errors of A2–A4.

Pooling those clips with the 11 validation clips that defeat A2–A5, we compared their recordings with all 1,260 validation and test clips (Table VII).

- *Recording problems*: the hard clips are far more often very quiet, contain several separate utterances, or have speech at the edge of the default 1.5 s window, which suggests the crop chose the wrong stretch. Two-thirds of them are very quiet, noisy or cropped at the edge, against 18% of all clips.

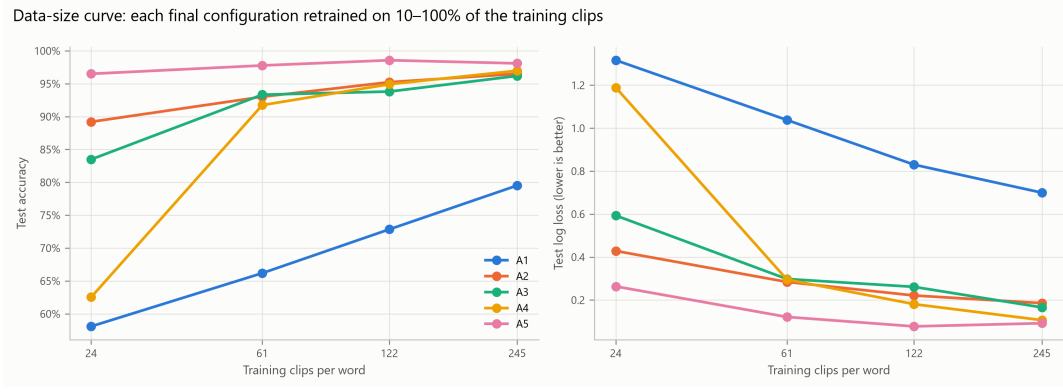


Fig. 4. Test accuracy and log loss when each final configuration is retrained on 10–100% of the training clips (24–245 per word; one seed per point).

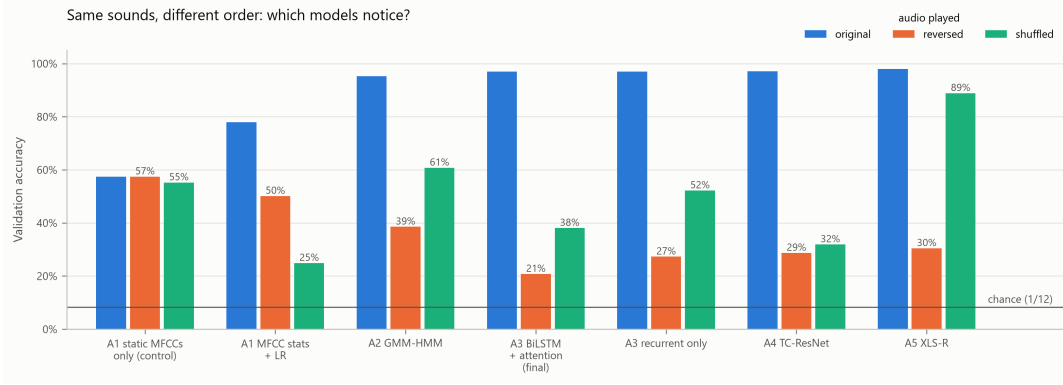


Fig. 5. Validation accuracy when each clip’s word is played unchanged, reversed, or as shuffled 100 ms chunks. The control is A1 refitted on static MFCC statistics only.

- *Spectrograms*: in about three of the ten hard test clips, the window shows no clear word at all, only clicks or near-silence.
- *Label-error candidates*: in three clips, A3–A5 agree on the same wrong word with at least 80% confidence. One is a test clip labelled *sita* that all four order-aware models hear as *tisa*, with 99–100% confidence.

Agreement among our models is not independent evidence of a label error, because they were all trained on these labels. We therefore checked the labels in two independent ways.

- *Confident learning* [42], applied to the out-of-sample probabilities of A3–A5, flags only 1 of the 1,260 validation and test clips: the *sita* clip above.
- *An external recogniser*: the multilingual MMS model with its Swahili adapter [22], which never saw this dataset, transcribed the 21 hard clips. On 60 easy test clips, it finds the labelled word in 82% and never names a wrong one, so its positive findings can be trusted.

The recogniser explains six of the hard clips.

- *One likely label error*: it hears *tisa* in the clip labelled *sita*, in agreement with confident learning and our models.
- *Three said in English*: two *hapana* clips contain “no”, and one *tisa* clip contains “nine”. Our models hear “nine” as *nane*.

- *Two crop failures*: the labelled word is in the recording, but outside the 1.5 s window.

The remaining transcripts contain other speech, such as a different Swahili phrase or English in the background, or nothing. Because the recogniser also misses the word in 18% of easy clips, these transcripts are weak evidence on their own. Six of the 21 hard clips are therefore clear collection problems. For most of the rest, the recording flags in Table VII point to difficult audio rather than to a weakness of one architecture. Excluding the two test clips that do not hold the labelled Swahili word raises every model’s test accuracy by 0.3 points and leaves the ranking unchanged. We did not relabel anything.

c) Noise is the main remaining risk: Error rates rise sharply in noisy recordings (Fig. 6).

- *SNR*: below 30 dB (135 test clips), A2–A5 misclassify 8–13% of clips, against at most 1.5% above 30 dB. A5 is the most robust (8.1%). It is also the only model trained with noise injection, but this study cannot separate that from its pretraining.
- *Loudness*: quiet recordings are harder for A2–A4 (4.7–8.3% errors) than for A5 (2.4%).
- *Duration*: words estimated to last over 0.7 s show higher error rates. Part of this is noise inflating the estimate: duration and SNR are correlated ($\rho = -0.42$), and among

TABLE VII
RECORDING FLAGS AMONG THE 21 CLIPS THAT EVERY ORDER-AWARE MODEL MISCLASSIFIES, AGAINST ALL 1,260 VALIDATION AND TEST CLIPS. THE WINDOW-EDGE FLAG USES THE DEFAULT 1.5 s WINDOW.

Flag	Hard clips	All clips
Very quiet (< -45 dBFS)	38%	4%
Several utterances	52%	25%
Speech at the window edge	43%	16%
Low SNR (< 15 dB)	14%	3%
Very short (< 0.3 s of speech)	10%	1%
Clipped	0%	4%

the 50 long clips with good SNR, A5 makes one error. A2–A4 still make two or three there (4–6%), above their overall rates.

d) What the models attend to: *nne* and *nane* differ in the first half of the word, where *nane* has the vowel /a/.

- A3: its attention peaks about a quarter of the way into both words. The profiles of the two words are indistinguishable ($p = 0.79$). A bidirectional LSTM summarises the whole word in every state, so the attention shows where the model reads its summary, not which sounds decide.
- A4: its class-activation evidence is spread over the whole word. For *nane* it shifts slightly towards the start (56% against 52% in the first half, $p = 0.014$), as expected if the vowel matters.
- *Early recognition:* a forward-only A3 that sees only the beginning of the word reaches 24% accuracy 100 ms after the onset, 80% at 500 ms and 88% at 800 ms, against 93.7% for the whole window (validation). The evidence separating the confusable words arrives with their second syllable.

Perturbation tests such as the stress test give firmer evidence about the use of order than these maps do.

B. Limitations

- 1) *Unknown speakers.* The dataset gives no speaker identities, so the frozen split cannot be speaker-disjoint. The same person may appear in both training and test clips, and every score may overstate performance on new speakers. Keyword-spotting benchmarks usually avoid this by keeping each speaker in one partition [7].
- 2) *A closed vocabulary.* Every clip is forced into one of 12 words, and there is no “unknown word” or “silence” class. A deployed system would confidently mislabel anything outside the vocabulary.
- 3) *One region, one collection.* All recordings come from crowd-workers in Kenya, collected once. The results may not transfer to other varieties of Swahili (for example in Tanzania or the DRC), to other recording devices, or to conversational speech.
- 4) *Augmentation constrained by the rules.* External data such as noise corpora is not allowed. Our synthetic noise (10–30 dB SNR) was harsher than most real clips

(median SNR 42 dB), so the augmentation study is only a partial test.

- 5) *Selection on one validation split.* Configurations were chosen from many validation runs, so validation scores are optimistic [41]. Differences of about one seed standard deviation (0.004 log loss for A3, 0.023 for A4) are indistinguishable from noise. The test split, never used for selection, gives the unbiased estimate.
- 6) *Early stopping.* A patience of five epochs stopped one seed each of A3 and A4 at epoch 16 of 40, while the learning rate was still high. For A4 that run explains most of the larger seed spread; A3’s seeds stayed close (0.004). We did not retune the patience after seeing the seed runs, to avoid selecting on validation once more.
- 7) *Label and collection noise.* 10 test and 11 validation clips defeat every order-aware model. Confident learning and an external recogniser identify one likely label error, three words said in English, and two recordings where the crop missed the word. For most of the rest the recogniser hears other speech or nothing, which is weak evidence on its own (Section VI). No one listened to the clips: our evidence is automatic, and the recogniser is itself fallible on short, noisy clips. We did not relabel anything, so these clips count as errors for every model. Removing the clear cases changes test accuracy by only 0.3 points.
- 8) *Pretraining confounds.* The two A5 backbones differ in size (316M against 95M parameters) as well as in pretraining language, and XLS-R’s pretraining included only 91 hours of Swahili. A5’s gains therefore cannot be attributed to Swahili data or to the language of pretraining alone. A Transformer trained from scratch, which would separate the architecture from the pretraining, was not run.
- 9) *A small test set.* With 630 test clips, accuracy is known to within about ± 1 –3 points, and each sound-alike pair has only about 105 clips.
- 10) *Sensitive metrics.* A few confident errors can dominate log loss, so it is always reported with accuracy and calibration error. ECE in turn depends on the binning, and its low-confidence bins are sparse.

C. Responsible AI

a) Consent and data protection: Voice recordings are biometric data. The clips were crowdsourced by the challenge organiser under its own terms, and we honoured its rules: the audio and derived features were never published, were shared only through private datasets among the group, and are excluded from the public repository.

b) Fairness we could not measure: The data carries no speaker metadata (gender, age, dialect or region), so we could not test whether error rates differ between groups. The wide pitch range suggests a mix of voices, but that is no substitute for a fairness audit. Speakers of other Swahili varieties, children and older adults may be served worse by any model trained only on these recordings.

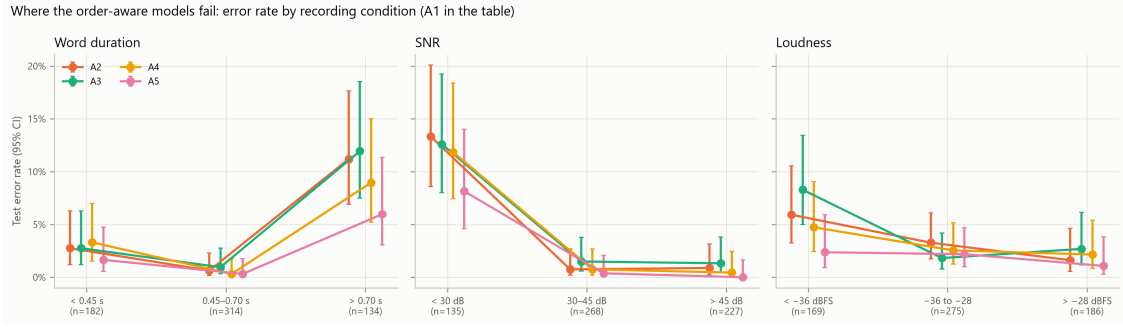


Fig. 6. Test error rate of the order-aware models by estimated word duration, SNR and loudness, with 95% Wilson intervals.

What we could measure is recording conditions, and they matter. In noisy recordings (SNR below 30 dB), the order-aware models make 8–13% errors, against at most 1.5% in cleaner ones (Section VI). Users with cheap phones or in noisy places, often the same users that spoken interfaces aim to include, would therefore see several times more errors than the headline accuracy suggests.

c) Consequences of errors: The intended applications, such as spoken digits for mobile money, helplines or menu navigation for users with limited literacy [1], turn a mis-recognised digit into a wrong transaction or a wrong service. Calibration therefore matters as much as accuracy. A deployed system should confirm uncertain inputs, which is why we report log loss and calibration error, and why we apply temperature scaling. Because the vocabulary is closed, it also needs a way to reject speech outside the vocabulary.

d) Dual use: Keyword spotting in under-resourced African languages has humanitarian uses, such as monitoring radio for relief programmes [24]. The same technology could also be used for surveillance of speech. The models here recognise only 12 isolated words, which limits that risk.

e) Environmental cost: The neural models were trained on one Kaggle Tesla T4, except one A4 run on a laptop CPU; the classical models ran on CPUs. All 19 logged A3 runs took about 10 minutes of GPU time in total. All of A4’s runs, including the failed and superseded ones, took under 30 minutes. The pretrained A5 dominates the budget: its 16 runs, including one failed run and the three data-size runs, took about 2.8 hours. That is about 11 minutes per run, against about one minute or less for A3 and A4. Even so, the compute and energy cost of the study is small, and we fine-tuned the smallest XLS-R model rather than the 1B or 2B versions.

f) Use of AI assistance: We used AI to plan the project and to help us understand some complex situations.

VII. CONCLUSION AND FUTURE WORK

We asked how effectively sequential models recognise spoken Swahili words. On 12 crowdsourced Swahili words they do so very effectively, and the evidence points to three findings and one limit.

- *Order:* the order-aware models reach 96–98% accuracy, against 79.5% for order-free statistics. The models also

differ in capacity, but the anagram pair and the stress test show that every order-aware model depends on order.

- *The kind of sequence model matters less than expected.* A classical HMM, a BiLSTM and a TC-ResNet reach similar accuracy. The convolutional model has the lowest log loss of the three (seed 42; within two seed standard deviations across seeds), with under a fifth of the BiLSTM’s parameters.
- *Pretraining* brings the largest gain we observed, although our design cannot separate it from A5’s architecture and size. The fine-tuned XLS-R is the best and most stable model (98.1% test accuracy, log loss 0.093). With 24 clips per word it is about as accurate as the models trained from scratch on ten times more data. The price is 96 times the latency and 2,100 times the size of A4.
- *The limit is the data.* The clips that defeat every order-aware model are mostly quiet, noisy or badly cropped, and include a likely label error and words said in English. Noise is the main risk for deployment.

a) Future work:

- *Speaker-disjoint evaluation:* cluster speaker embeddings into pseudo-speakers, to measure how much speaker overlap inflates our scores.
- *Noise robustness:* multi-condition training with realistic noise, where the rules allow it, since errors concentrate below 30 dB SNR.
- *Rejecting other speech:* an “unknown” and “silence” class, and voice-activity detection, to avoid the crop failures and to reject words outside the vocabulary.
- *Separating architecture from pretraining:* a Transformer trained from scratch [17], and a backbone covering more African languages (MMS [22]).
- *Deployment:* distilling A5 into an A4-sized model for phones, and streaming recognition, for which a forward-only recurrent model already reaches 80% accuracy 500 ms into the word.

TEAM AND CONTRIBUTIONS

Ownership followed the project plan: Hassan Adelani Luqman (M1) the data pipeline, evaluation and baselines A1–A2; Haguma Bianca (M2) augmentation, the training loop and A3; Emmanuel Mukasa (M3) the EDA, A4 and the error analysis;

Emmanuel Nsabagasani (M4) A5 and the final evaluation. Commits and tasks per member are in the contribution tracker.

AI-USE STATEMENT

AI was used extensively to plan the project, to understand complex scenarios, and to help write some of the codes.

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